

Practical Laboratory Audit & Governance

Report

Course	GRC102 – Information Security Governance
Module	Module 4 – Monitoring and Auditing Security Controls
Week	Week 4 (26 September – 02 October 2026)
Submission Due	02 October 2026, 11:59 PM WAT
Student Name	Zainab Umar Shuaibu
Assigned Lab	ICDFA Linux Practice Hub (icdfa@icdfa)

GRC102 – Information Security Governance | Week 4 Lab Report Page 1

Executive Summary

An empirical security monitoring, log analysis, and system hardening assessment was conducted on the assigned Linux workload (icdfa@icdfa). Using technical evidence captured directly from `auditd`, `systemd-journald`, and Lynis 3.1.4, host-level events were evaluated against enterprise risk and control objectives.

Top 3 Key Governance Findings

- **Authentication Log Relocation & Legacy Path Failure:** Standard log auditing at `/var/log/auth.log` failed because the legacy file was missing. System authentication is managed through `systemd-journald`, requiring updated ingestion pipelines for enterprise visibility.
- **Active Remote Authentication Failures (SSH Brute Force Simulation):** Live journal monitoring (`journalctl -f -u ssh`) captured multiple failed authentication attempts for invalid user `wronguser` via `pam_unix(sshd:auth)`.
- **Hardening Deficits & Unencrypted Storage:** Initial Lynis execution revealed unencrypted partitions (`/dev/sda1`), missing brute-force protections (`fail2ban`), unconfigured password protection on GRUB, and missing auxiliary security packages (`needrestart`, `apt-listbugs`).

1. Scope, Authorisation & Methodology

- **Scope & Boundaries:** Testing was conducted exclusively within the assigned isolated VirtualBox environment (icdfa@icdfa) running Kali Linux Rolling Release (Kernel 6.12.13). No external network targets were accessed.
- **System Auditing:** Verification and rule configuration using `auditd` and `auditctl`.
- **Log Management:** Forensic analysis of system authentication events using `systemd-journald` /

journalctl. • **Security Assessment:** Automated configuration and vulnerability scanning using Lynis 3.1.4.

- **Governance Mapping:** Alignment of observed technical evidence with control objectives, ownership, KPIs/KRIs, and escalation workflows.

2. Module 1: System Auditing with auditd

Activity 1.1 — Service Status & Activation

The audit daemon was enabled and initialized via systemd:

```
$ sudo systemctl start auditd
$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install. Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' -> '/usr/lib/systemd/system/auditd.service'.
```

Activity 1.2 — Loaded Custom Audit Rules (/etc/audit/rules.d/custom.rules) To

maintain non-repudiation and meet regulatory compliance requirements, the following rules were configured:

```
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
```

Evidence Analysis & Governance Interpretation

- **Evidence Collected:** System service symlinks created and auditd active.
- **Interpretation:** Enabling kernel-level tracking provides non-repudiable logs of file modifications and binary executions. This satisfies ISO/IEC 27001 A.12.4.1 and NIST CSF DE.CM requirements.

3. Module 2: Log Management and Analysis

Activity 2.1 & 2.2 — Authentication Evidence & Findings

GRC102 – Information Security Governance | Week 4 Lab Report Page 2

1. **Legacy File Inspection Failure:** Executing `sudo grep -i "failed" /var/log/auth.log` returned `grep: /var/log/auth.log: No such file or directory`.

2. **Journalctl Analysis:** Starting the SSH daemon and following the journal captured an active login test:

```
Oct 02 08:16:49 icdfa systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 02 08:16:50 icdfa sshd[7701]: Server listening on 0.0.0.0 port 22.
Oct 02 08:18:22 icdfa sshd-session[8268]: Invalid user wronguser from 127.0.0.1 port 48814
Oct 02 08:18:27 icdfa sshd-session[8268]: pam_unix(sshd:auth): check pass; user unknown
Oct 02 08:18:27 icdfa sshd-session[8268]: pam_unix(sshd:auth): authentication failure; logname= uid=0
euid=0 tty=ssh ruseOct 02 08:18:29 icdfa sshd-session[8268]: Failed password for invalid user
wronguser from 127.0.0.1 port 48814 ssh2 Oct 02 08:18:49 icdfa sshd-session[8268]: PAM 2 more
authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=
```

Authentication Event Table

Event / Condition	Evidence Source	Time (WAT)	User / Service	Security Significance	Recommended Action
Invalid User Access	journalctl -f -u ssh	Oct 02 08:18:22	wronguser / sshd		enumeration attempt. Disable SSH keys
PAM Auth Failure	journalctl -f -u ssh	Oct 02 08:18:27	uid=0 / pam_unix		word attempts logged by PAM. y block offending IPs.
Missing Log File	grep /var/log/auth.log	Oct 02 08:16:00	System / rsyslog		ay fail without systemd-journal e SIEM agents to read system

4. Module 3: Security Assessment with Lynis

- **Tool Version:** Lynis 3.1.4 (CISOfy)
- **Kernel Version:** Linux 6.12.13 (Kali Rolling Release x86_64)
- **Target Host:** icdffa

```
[WARNING]: Test DEB-0001 had a long execution: 37.383094 seconds
- libpam-tmpdir [ Not Installed ]
[WARNING]: Test DEB-0280 had a long execution: 17.211419 seconds
- Checking / on /dev/sda1 [ NOT ENCRYPTED ]
- Software:
- fail2ban [ Not Installed ]
- needrestart [ Not Installed ]
- GRUB Password Protection [ NONE ]
```

Finding ID	Identified Issue / Risk	Evidence Captured	Control Owner	Recommended Remediation	Priority
LYN-01	Unencrypted Root Storage	/dev/sda1 [NOT ENCRYPTED]	Security Operations	Implement LUKS encryption on primary boot partition.	High
LYN-02	Missing Brute-Force Protection	fail2ban not installed		Install and configure fail2ban to drop repeat SSH login attempts.	Medium
LYN-03	Unprotected Bootloader	GRUB password [NONE]	System Administrator	Set password hash in /etc/grub.d/ to prevent unauthorized boot.	Medium
LYN-04	Missing Temp Isolation	libpam-tmpdir not installed	System Administrator	Install libpam-tmpdir to isolate per-user directories.	Medium

Objective	Technical Evidence	Owner	Status	KPI / KRI	Threshold	Risk Significance	Remediation	Retest / Follow-up	File Integrity	auditd rules (passwd_changes)	IT
SecOps Compliant	0 unauthorized edits (Target: 0)	Privilege escalation and unauthorized user creation. Maintain active audit rules on account files. Weekly ausearch querye-Force Defense journalctl PAM failures & Lynis SecOps / Network Deficient < 3 failures/min (Threshold: 5) Account guessing and password spraying attacks. Install fail2ban and set SSH key-only auth. Re-attempt failed login; metry Availability Missing /var/log/auth.log Monitoring Team Deficient 100% journal source ingestion (Target: 100%) Blind spots in security monitoring and loss of audit evidence. Map log collectors to systemd-journal socket. Verify real-time event pi	torage Security Lynis /dev/sda1 check Storage / IT Ops Deficient Unencrypted volumes (Target: 0) Physical theft or unauthorized offline drive mounting. Enforce LUKS full-disk encryption on new builds. Re-run Lynis storage cher Protection Lynis GRUB grub check SysAdmin Deficient Unprotected bootloaders (Target: 0) Local physical access exploitation and root password reset. Enable GRUB PBKDF2 authentication. Inspect /boot/grub/grub.								

[illegible]

Escalation Criteria

- **Operational Alerts (SOC Level 1):** Transient authentication errors, routine elevated privilege usage (`sudo`), or single daemon restarts.
- **Governance Escalations (CISO / Audit Committee):** High-frequency SSH brute-force attempts from internal networks, failure of centralized logging pipelines, or unauthorized modification of system configuration files.

7. Conclusion & Next Steps

This laboratory demonstrated the operational link between technical Linux auditing tools and enterprise security governance. By moving from static file checks to real-time telemetry monitoring via `systemd-journald` and `auditd`, security teams maintain verifiable evidence necessary to satisfy compliance frameworks such as ISO 27001 and NIST CSF. Immediate priorities include deploying `fail2ban`, enabling LUKS disk encryption, and re-routing SIEM ingestion to native `systemd` sockets.

Appendix: Evidence Bundles (Terminal Screenshots)

The following evidence bundles refer to screenshots captured during the laboratory. The original screenshot files were not included in the supplied text, so their filenames are retained as references rather than embedded images.

Evidence Bundle 1: Auditd Service Enabling & Missing Log

Discovery Referenced screenshot: Screenshot 2026-10-02 081723.png

Shows successful enabling of `auditd.service`, the missing `/var/log/auth.log` path, and SSH service initialization.

Evidence Bundle 2: SSH Brute Force Test Execution

Referenced screenshot: Screenshot 2026-10-02 081858.png

Shows execution of `ssh wronguser@127.0.0.1` generating password prompts and permission denials.

Evidence Bundle 3: Real-Time Journalctl Telemetry Capture

Referenced screenshot: Screenshot 2026-10-02 081919.png

Captures live SSH authentication failure logs, PAM user checks, and invalid user warnings in `systemd-journald`.

Evidence Bundle 4: Lynis 3.1.4 Initialization & Execution

Referenced screenshot: Screenshot 2026-10-02 082106.png & Screenshot 2026-10-02

082121.png Demonstrates Lynis audit start on the `icdfa` host running Kali Linux Rolling release.

Evidence Bundle 5: Lynis System Hardening Warnings & Software Check

Referenced screenshot: Screenshot 2026-10-02 082154.png

Shows explicit findings including unencrypted `/dev/sda1`, missing `fail2ban`, missing `libpam-tmpdir`, and unconfigured GRUB password.

File Actions Edit View Help

```
Processing triggers for libc-bin (2.41-6) ...
Processing triggers for man-db (2.13.0-1) ...
Processing triggers for kali-menu (2025.1.1) ...
```

```
(icdfa@icdfa)-[~]
$ sudo systemctl start auditd
```

```
(icdfa@icdfa)-[~]
$ sudo systemctl enable auditd
```

```
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' → '/usr/lib/systemd/system/auditd.service'.
```

```
(icdfa@icdfa)-[~]
$ sudo grep -i "failed" /var/log/auth.log
```

```
grep: /var/log/auth.log: No such file or directory
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl -u ssh | grep -i "failed"
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl | grep -i "failed password"
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl -f -u ssh
```

```
^C
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl -u ssh --no-pager
```

```
-- No entries --
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl -u sshd --no-pager
```

```
-- No entries --
```

```
(icdfa@icdfa)-[~]
$ sudo systemctl start ssh
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl -f -u ssh
```

```
Oct 02 08:16:49 icdfa systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 02 08:16:50 icdfa sshd[7701]: Server listening on 0.0.0.0 port 22.
Oct 02 08:16:50 icdfa sshd[7701]: Server listening on :: port 22.
Oct 02 08:16:50 icdfa systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

File Actions Edit View Help

```
zsh: corrupt history file /home/icdfa/.zsh_history
```

```
(icdfa@icdfa)-[~]
```

```
$ ssh wronguser@127.0.0.1
```

```
The authenticity of host '127.0.0.1 (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:Kv90xH3mBWqri60rbMcRfC/t5u840J0Btwnb6gtSy1M.
This key is not known by any other names.
```

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
```

```
Please type 'yes', 'no' or the fingerprint: yes
```

```
Warning: Permanently added '127.0.0.1' (ED25519) to the list of known hosts.
```

```
wronguser@127.0.0.1's password:
```

```
Permission denied, please try again.
```

```
wronguser@127.0.0.1's password:
```

```
Permission denied, please try again.
```

```
wronguser@127.0.0.1's password:
```

```
wronguser@127.0.0.1: Permission denied (publickey,password).
```

```
(icdfa@icdfa)-[~]
```

```
$
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl -f -u ssh
Oct 02 08:16:49 icdfa systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 02 08:16:50 icdfa sshd[7701]: Server listening on 0.0.0.0 port 22.
Oct 02 08:16:50 icdfa sshd[7701]: Server listening on :: port 22.
Oct 02 08:16:50 icdfa systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Oct 02 08:18:22 icdfa sshd-session[8268]: Invalid user wronguser from 127.0.0.1 port 48814
Oct 02 08:18:27 icdfa sshd-session[8268]: pam_unix(sshd:auth): check pass; user unknown
Oct 02 08:18:27 icdfa sshd-session[8268]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Oct 02 08:18:29 icdfa sshd-session[8268]: Failed password for invalid user wronguser from 127.0.0.1 port 48814 ssh2
Oct 02 08:18:39 icdfa sshd-session[8268]: pam_unix(sshd:auth): check pass; user unknown
Oct 02 08:18:41 icdfa sshd-session[8268]: Failed password for invalid user wronguser from 127.0.0.1 port 48814 ssh2
Oct 02 08:18:46 icdfa sshd-session[8268]: pam_unix(sshd:auth): check pass; user unknown
Oct 02 08:18:48 icdfa sshd-session[8268]: Failed password for invalid user wronguser from 127.0.0.1 port 48814 ssh2
Oct 02 08:18:49 icdfa sshd-session[8268]: Connection closed by invalid user wronguser 127.0.0.1 port 48814 [preauth]
Oct 02 08:18:49 icdfa sshd-session[8268]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
```

File Actions Edit View Help

```
(icdfa@icdfa)-[~]
$ sudo lynis audit system
```

[Lynis 3.1.4]

```
#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.
```

2007-2024, CISOfy - <https://cisofy.com/lynis/>
Enterprise support available (compliance, plugins, interface and tools)

```
#####
Trash
```

[+] Initializing program

```
- Detecting OS ... [ DONE ]
- Checking profiles ... [ DONE ]
```

```
Program version:      3.1.4
Operating system:     Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version:       6.12.13
Hardware platform:    x86_64
Hostname:             icdfa
```

```
Profiles:             /etc/lynis/default.prf
Log file:              /var/log/lynis.log
Report file:           /var/log/lynis-report.dat
Report version:        1.0
Plugin directory:      /etc/lynis/plugins
```

```
Auditor:              [Not Specified]
Language:              en
Test category:         all
Test group:            all
```

```
- Program update status ... [ NO UPDATE ]
```

[+] System tools

```
- Scanning available tools...
```

[+] **Initializing program**

- Detecting OS... [DONE]
- Checking profiles ... [DONE]

Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.13
Hardware platform: x86_64
Hostname: icdfa

Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins

Auditor: [Not Specified]
Language: en
Test category: all
Test group: all

- Program update status ... [NO UPDATE]

[+] **System tools**

-
- Scanning available tools ...
 - Checking system binaries ...

[+] **Plugins (phase 1)**

Note: plugins have more extensive tests and may take several minutes to complete

- Plugin: **debian**
[

[+] **Debian Tests**

-
- Checking for system binaries that are required by Debian Tests ...
 - Checking /bin ... [FOUND]
 - Checking /sbin ... [FOUND]
 - Checking /usr/bin ... [FOUND]

[+] Plugins (phase 1)

Note: plugins have more extensive tests and may take several minutes to complete

- Plugin: debian

[

[+] Debian Tests

- Checking for system binaries that are required by Debian Tests ... [FOUND]
- Checking /bin ... [FOUND]
- Checking /sbin ... [FOUND]
- Checking /usr/bin ... [FOUND]
- Checking /usr/sbin ... [FOUND]
- Checking /usr/local/bin ... [FOUND]
- Checking /usr/local/sbin ... [FOUND]
- Authentication:
- PAM (Pluggable Authentication Modules):

[WARNING]: Test DEB-0001 had a long execution: 37.383094 seconds

- libpam-tmpdir [Not Installed]
- File System Checks:
- DM-Crypt, Cryptsetup & Cryptmount:

[WARNING]: Test DEB-0280 had a long execution: 17.211419 seconds

- Checking / on /dev/sda1 [NOT ENCRYPTED]
- Software:
- apt-listbugs [Not Installed]
- apt-listchanges [Not Installed]
- needrestart [Not Installed]
- fail2ban [Not Installed]

]

[+] Boot and services

- Service Manager [systemd]
- Checking UEFI boot [DISABLED]
- Checking presence GRUB2 [FOUND]
- Checking for password protection [NONE]
- Check running services (systemctl) [DONE]
Result: found 20 running services
- Check enabled services at boot (systemctl) [DONE]
Result: found 20 enabled services

